MOBSF

ANDROID STATIC ANALYSIS REPORT



Diva (1.0)

File Name:	diva-app-debug.apk
Package Name:	jakhar.aseem.diva
Scan Date:	Oct. 28, 2026, 6:02 p.m.
App Security Score:	47/100 (MEDIUM RISK)
Grade:	B

FINDINGS SEVERITY

HIGH	MEDIUM	INFO	SECURE	HOTSPOT
2	8	1	1	1

FILE INFORMATION

File Name: diva-app-debug.apk
Size: 1.38MB
MD5: 299b15c63aac6aae44ce42281e768b4c
SHA1: 343a4f2214091414c6566ca00b0963dc421a4453
SHA256: 4adc2de8a86d958fce181a067588fa9891f67be58b87aaf0629750ee9fa28331

APP INFORMATION

App Name: Diva
Package Name: jakhar.aseem.diva
Main Activity: .MainActivity
Target SDK: 23
Min SDK: 15
Max SDK:
Android Version Name: 1.0
Android Version Code: 1

APP COMPONENTS

Activities: 17
Services: 0
Receivers: 0
Providers: 1
Exported Activities: 2
Exported Services: 0
Exported Receivers: 0
Exported Providers: 1

CERTIFICATE INFORMATION

Binary is signed
v1 signature: True
v2 signature: True
v3 signature: True
v4 signature: False
v3.1 signature: False
v3.2 signature: False
X.509 Subject: C=US, O=Android, CN=Android Debug
Signature Algorithm: rsassa_pkcs1v15
Valid From: 2026-10-28 17:57:54+00:00
Valid To: 2054-03-15 17:57:54+00:00
Issuer: C=US, O=Android, CN=Android Debug
Serial Number: 0xd77844b9e579e806
Hash Algorithm: sha384
md5: fde4bf66a2857760ecb1198eaa2a8ee5
sha1: 3e1efe30cf27b28b0d0b41764f2f844753d9934f
sha256: 8c69cc1a8d626dc415fe31b4cd82bf40cc05b58f95312ef7ea52ffe316173f2f
sha512: 32b6486e36b3c82bc37ad5e505943f2eb999749b8035c1eaf163ae699d6898f924e2f341e90236a5bc386998dd2a55cb665204f6c7fdf2495387f6379c3aa864
PublicKey Algorithm: rsa
Bit Size: 2048
Fingerprint: 14d3c5e2855c9815335b42c692214ca7d083c43fe71e47d53f1d8345f21ff5a1
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.WRITE_EXTERNAL_STORAGE	dangerous	read/modify/delete external storage contents	Allows an application to write to external storage.
android.permission.READ_EXTERNAL_STORAGE	dangerous	read external storage contents	Allows an application to read from external storage.
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.

APKID ANALYSIS

FILE	DETAILS
classes.dex	FINDINGS: Compiler — DETAILS: r8

NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION

CERTIFICATE ANALYSIS

HIGH: 1 | WARNING: 1 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application vulnerable to Janus Vulnerability	warning	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.
Application signed with debug certificate	high	Application signed with a debug certificate. Production application must not be shipped with a debug certificate.

MANIFEST ANALYSIS

HIGH: 1 | WARNING: 4 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 4.0.3-4.0.4, [minSdk=15]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
3	Activity (.APICredsActivity) is not Protected. An intent-filter exists.	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Activity is explicitly exported.

NO	ISSUE	SEVERITY	DESCRIPTION
4	Activity (.APICreds2Activity) is not Protected. An intent-filter exists.	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Activity is explicitly exported.
5	Content Provider (.NotesProvider) is not Protected. [android:exported=true]	warning	A Content Provider is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.

</> CODE ANALYSIS

HIGH: **0** | WARNING: **3** | INFO: **1** | SECURE: **0** | SUPPRESSED: **0**

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	The App logs information. Sensitive information should never be logged.	info	CWE: CWE-532: Insertion of Sensitive Information into Log File OWASP MASVS: MSTG-STORAGE-3	jakhar/aseem/diva/AccessControl1Activity.java jakhar/aseem/diva/AccessControl2Activity.java jakhar/aseem/diva/InsecureDataStorage2Activity.java jakhar/aseem/diva/InsecureDataStorage3Activity.java jakhar/aseem/diva/InsecureDataStorage4Activity.java jakhar/aseem/diva/LogActivity.java jakhar/aseem/diva/SQLInjectionActivity.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
2	App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database.	warning	CWE: CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') OWASP Top 10: M7: Client Code Quality	jakhar/aseem/diva/InsecureDataStorage2Activity.java jakhar/aseem/diva/NotesProvider.java jakhar/aseem/diva/SQLInjectionActivity.java
3	App creates temp file. Sensitive information should never be written into a temp file.	warning	CWE: CWE-276: Incorrect Default Permissions OWASP Top 10: M2: Insecure Data Storage OWASP MASVS: MSTG-STORAGE-2	jakhar/aseem/diva/InsecureDataStorage3Activity.java
4	App can read/write to External Storage. Any App can read data written to External Storage.	warning	CWE: CWE-276: Incorrect Default Permissions OWASP Top 10: M2: Insecure Data Storage OWASP MASVS: MSTG-STORAGE-2	jakhar/aseem/diva/InsecureDataStorage4Activity.java

SHARED LIBRARY BINARY ANALYSIS

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
1	armeabi-v7a/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
2	arm64-v8a/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
3	riscv64/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
4	x86/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
5	x86_64/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
6	armeabi-v7a/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
7	arm64-v8a/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
8	riscv64/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
9	x86/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
10	x86_64/libdivajni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__strcpy_chk']	True info Symbols are stripped.

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION

BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00022	Open a file from given absolute path of the file	file	jakhar/aseem/diva/InsecureDataStorage4Activity.java

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	3/25	android.permission.WRITE_EXTERNAL_STORAGE, android.permission.READ_EXTERNAL_STORAGE, android.permission.INTERNET
Other Common Permissions	0/44	

Malware Permissions:
Top permissions that are widely abused by known malware.
Other Common Permissions:
Permissions that are commonly abused by known malware.

! OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION

DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION
payatu.com	ok	No Geolocation information available.

HARDCODED SECRETS

POSSIBLE SECRETS
"pkey" : "notespin"

SCAN LOGS

Timestamp	Event	Error
2026-10-28 18:02:33	Generating Hashes	OK

2026-10-28 18:02:33	Extracting APK	OK
2026-10-28 18:02:33	Unzipping	OK
2026-10-28 18:02:34	Parsing APK with androguard	OK
2026-10-28 18:02:34	Extracting APK features using aapt/aapt2	OK
2026-10-28 18:02:34	Getting Hardcoded Certificates/Keystores	OK
2026-10-28 18:02:37	Parsing AndroidManifest.xml	OK
2026-10-28 18:02:37	Extracting Manifest Data	OK
2026-10-28 18:02:37	Manifest Analysis Started	OK
2026-10-28 18:02:37	Performing Static Analysis on: Diva (jakhar.aseem.diva)	OK
2026-10-28 18:02:37	Checking for Malware Permissions	OK

2026-10-28 18:02:37	Fetching icon path	OK
2026-10-28 18:02:37	Library Binary Analysis Started	OK
2026-10-28 18:02:37	Analyzing lib/armeabi-v7a/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing lib/arm64-v8a/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing lib/riscv64/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing lib/x86/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing lib/x86_64/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing apktool_out/lib/armeabi-v7a/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing apktool_out/lib/arm64-v8a/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing apktool_out/lib/riscv64/libdivajni.so	OK
2026-10-28 18:02:37	Analyzing apktool_out/lib/x86/libdivajni.so	OK

2026-10-28 18:02:37	Analyzing apktool_out/lib/x86_64/libdivajni.so	OK
2026-10-28 18:02:37	Reading Code Signing Certificate	OK
2026-10-28 18:02:37	Running APKiD 3.2.0	OK
2026-10-28 18:02:38	Detecting Trackers	OK
2026-10-28 18:02:38	Decompiling APK to Java with JADX	OK
2026-10-28 18:02:47	Converting DEX to Smali	OK
2026-10-28 18:02:47	Code Analysis Started on - java_source	OK
2026-10-28 18:02:47	Android SBOM Analysis Completed	OK
2026-10-28 18:02:47	Android SAST Completed	OK
2026-10-28 18:02:47	Android API Analysis Started	OK
2026-10-28 18:02:48	Android API Analysis Completed	OK

2026-10-28 18:02:48	Android Permission Mapping Started	OK
2026-10-28 18:02:49	Android Permission Mapping Completed	OK
2026-10-28 18:02:49	Android Behaviour Analysis Started	OK
2026-10-28 18:02:50	Android Behaviour Analysis Completed	OK
2026-10-28 18:02:50	Extracting Emails and URLs from Source Code	OK
2026-10-28 18:02:50	Email and URL Extraction Completed	OK
2026-10-28 18:02:50	Extracting String data from APK	OK
2026-10-28 18:02:50	Extracting String data from SO	OK
2026-10-28 18:02:50	Extracting String data from Code	OK
2026-10-28 18:02:50	Extracting String values and entropies from Code	OK
2026-10-28 18:02:50	Performing Malware check on extracted domains	OK

2026-10-28 18:02:50	Saving to Database	OK

Report Generated by - MobSF v4.6.2

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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